



DP-602: Implement a data warehouse with Microsoft Fabric

Exercise 5 : Secure data in a data warehouse

- Create a workspace
- Create a data warehouse
- Apply dynamic data masking rules to columns in a table
- Apply row-level security
- Implement column-level security
- Configure SQL granular permissions using T-SQL



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➤ Create a workspace

Create a workspace

✕

Name *

secure_DW

✔ This name is available

Description

Describe this workspace

Domain ⓘ

Assign to a domain (optional) ▾

[Learn more about workspace settings](#) [↗](#)

Workspace image

[↑ Upload](#)
[↶ Reset](#)

Advanced ^

Contact list * ⓘ

dagstudentsgreatlearning (Owner) ✕

Workspace type ⓘ

What a user can do in a workspace depends on the individual user license(s) assigned to that user and the workspace type. [Learn more](#) [↗](#)

Power BI only workspace types

Power BI workspaces only support the use of Power BI capabilities.

Apply

Cancel



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Create a data warehouse

New Warehouse

Name *

Location *

MS Fabric Practice

▼

Advanced

Create

Cancel

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Management

Help

Analyze data with ▼

Share

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Copilot

Create

Browse

Apps

Scorecards

OneLake catalog

Learn

Workspaces

MS Fabric Practice

secure_DW

...

Power BI

Explorer

Warehouses

secure_DW

Schemas

Security

Queries

My queries

My queries

Shared queries

Build a warehouse

Start getting data

New Copy job

Keep warehouse tables in sync with scheduled data transfers

New Dataflow Gen2

Ingest and transform data with a low-code interface.

Practice with sample data

Automatically load sample data in your warehouse.

Start developing

T-SQL

Create tables with T-SQL code templates.

SQL database project

Publish a database schema from an external project.

Need help? [Learn more](#)



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Apply dynamic data masking rules to columns in a table

There are four types of masks:
default, email, random and custom string

```
1  
2  
3  
4  
5 CREATE TABLE dbo.Customers  
6 (  
7     CustomerID INT NOT NULL,  
8     FirstName varchar(50) MASKED WITH (FUNCTION='Partial(1,"XXXXXXX",0)') NULL,  
9     LastName varchar(50) NOT NULL ,  
10    Phone varchar(20) MASKED WITH (FUNCTION='default()') NULL,  
11    Email varchar(50) MASKED WITH (FUNCTION='email()') NULL  
12 );  
13  
14 insert dbo.Customers (CustomerID , FirstName, LastName, Phone,Email) VALUES  
15 (29485,'Catherine','Abel','555-555-5555','catherine0@adventure-works.com'),  
16 (29486,'Kim','Abercrombie','444-444-4444','Kim2@adventure-works.com'),  
17 (29489,'Frances','Adams','333-333-333','frances0@adventure-works.com');  
18  
19 select * from dbo.Customers;
```

123 CustomerID	ABC FirstName	ABC LastName	ABC Phone	ABC Email
29485	CXXXXXXX	Abel	xxxx	cXXX@XXXX.com
29486	KXXXXXXX	Abercrombie	xxxx	kXXX@XXXX.com
29489	FXXXXXXX	Adams	xxxx	fXXX@XXXX.com



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Apply row-level security

SQL query 1

Run Save as Save as view Create rule Explain query Fix query errors

Preview Copilot uses AI. Mistakes can happen. Verify code suggestions before running them. [Review terms](#)

```
1  
2  
3 select * FROM dbo.Sales;
```

Messages Results

	123	OrderID	ABC SalesRep	ABC Product	123	Quantity
1	1		<username1>@<your_domain>.com	Valve		5
2	2		<username1>@<your_domain>.com	Wheel		2
3	3		<username1>@<your_domain>.com	Valve		4
4	4		<username2>@<your_domain>.com	Bracket		2
5	5		<username2>@<your_domain>.com	Wheel		5
6	6		<username2>@<your_domain>.com	Seat		5

mydatawarehouse

Search

Trial: 7 days left

Home Reporting Management Help

Get data New SQL query SQL templates Query activity Model layouts Download SQL database project Open in Copilot

Some features are unavailable because of your permissions or workspace role. To make these features available, have an admin change your permissions or workspace role. [Learn more](#)

Explorer

Warehouses

mydatawarehouse

Schemas

dbo

Tables

Custom Sales Views Functions Stored Pr...

INFORMATIO... queryinsights rls sys

SQL query 1

Run Save as view Explain query Fix query errors

```
1 SELECT * FROM dbo.Sales;
```

Messages Results

	123	OrderID	ABC SalesRep	ABC Product	123	Quantity
1	1		username1@your_domain.com	Valve		5
2	2		username1@your_domain.com	Wheel		2
3	3		username1@your_domain.com	Valve		4

Succeeded (5 sec 389 ms) Copilot completions: Off

Columns: 4 Rows: 3



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► Implement column-level security

The screenshot displays the Microsoft Fabric SQL interface. On the left, the 'Explorer' pane shows the 'secure_DW' warehouse structure, including schemas (dbo, Sales, Views, Functions, Stored Procedures, INFORMATION_SCHEMA, queryinsights) and tables (Customers, Orders). The 'Orders' table is expanded, showing columns: orderID (int, null), CustomerID (int, null), and CreditCard (varchar(20), null). The main pane shows 'SQL query 3' with the following T-SQL statement:

```
1  
2 DENY SELECT ON dbo.Orders (CreditCard) TO [<username1>@<your_domain>.com];
```

Below the query editor, the 'Messages' pane shows the execution results:

Time	Message
11:51:04 PM	Started executing on line 1 Msg 33130, Level 16, State 1, Line 1 Principal '<username1>@<your_domain>.com' could not be found or this principal type is not supported.
11:51:06 PM	Query execution time: 00:00:01.200 Time to process and render results: 00:00:00.761 Total duration: 00:00:01.961

Deny permission to view a column in the table. The T-SQL statement prevents <username1>@<your_domain>.com from seeing the CreditCard column in the Orders table. In the DENY statement, replace <username1>@<your_domain>.com with the user name of a user who has Viewer permissions on the workspace.

code



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► Configure SQL granular permissions using T-SQL

The screenshot shows the Microsoft Fabric SQL interface. On the left is the 'Explorer' pane with a tree view of the 'secure_DW' warehouse, including schemas (dbo, INFORMATION_SCHEMA, queryinsights, rls), tables (Customers, Orders, Parts, Sales), views, functions, and stored procedures. The main editor displays 'SQL query 4' with the following T-SQL code:

```
1 CREATE PROCEDURE dbo.sp_printMessage
2 AS
3 PRINT 'Hello World';
4 GO
5 CREATE TABLE dbo.Parts
6 (
7     PartID INT ,
8     PartName VARCHAR(25)
9 );
10
11
12 INSERT dbo.Parts (PartID, PartName) VALUES
13 (1234, 'Wheel'),
14 (5678, 'Seat');
15
16 GO
17
18
19 EXEC dbo.sp_printMessage;
20
21 GO
22 select * from dbo.Parts;
```

Below the editor, the 'Results' tab is active, displaying a table with two columns: 'PartID' and 'PartName'. The table contains two rows of data:

PartID	PartName
1234	Wheel
5678	Seat

The screenshot shows the Microsoft Fabric SQL interface with 'SQL query 3' selected. The query is:

```
1 DENY SELECT on dbo.Parts to [<username1>@<your_domain>.com];
2
3 GRANT EXECUTE on dbo.sp_PrintMessage to [<username1>@<your_domain>.com];
```

The 'Messages' pane at the bottom shows the execution results:

12:09:29 AM Started executing on line 1
Msg 33130, Level 16, State 1, Line 1
Principal '<username1>@<your_domain>.com' could not be found or this principal type is not supported.

12:09:31 AM Query execution time: 00:00:00.865 | Time to process and render results: 00:00:01.238 | Total duration: 00:00:02.103



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Reference

